

MD. ISMAIL HOSSAIN

DevOps Engineer

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PROFESSIONAL SUMMARY

DevOps Engineer with hands-on experience running production infrastructure across Kubernetes, high-availability databases, observability, and virtualization. Currently at Onnorokom Projukti, operating multi-node HA K3s clusters, self-managed Redis/ClickHouse/MongoDB/PostgreSQL, GitOps delivery with ArgoCD, and end-to-end monitoring (Prometheus, Grafana, SigNoz, ELK). Proven record of cutting database failover downtime ~10x (approx. 5s to ~500ms), migrating multi-million-record datastores with zero data loss, and eliminating Elasticsearch mapping explosions (7,600+ to ~25 fields) that removed multi-minute Kibana load times. Also building self-hosted AI-Ops agents (LLM-driven log/metric anomaly detection and automated root-cause analysis). Backed by a strong Linux/Windows system-administration and networking foundation.

CORE COMPETENCIES

Kubernetes, K3s, Docker, Helm, ArgoCD, GitOps, CI/CD, Terraform, Ansible, Jenkins, GitLab, Prometheus, Grafana, SigNoz, OpenTelemetry, Elasticsearch, ELK, Redis, PostgreSQL, MongoDB, ClickHouse, Kafka, RabbitMQ, Ceph, MinIO, Proxmox, VMware, Hyper-V, CloudStack, AWS, Linux, Bash, Python, HAProxy, Keepalived, Istio, Envoy, MetalLB, cert-manager, AIOps, LLM-Ops, kagent, HolmesGPT, KRR, Drain3, Log Analytics, Anomaly Detection, High Availability, Disaster Recovery, Observability, Infrastructure as Code, System Administration.

TECHNICAL SKILLS

Containers & Orchestration: Docker, Docker Compose, Kubernetes (K3s), Helm, Kustomize, ArgoCD (GitOps), Portainer

Ingress, Mesh & Load Balancing: Envoy Gateway (Gateway API), Istio, MetalLB, NGINX / NGINX Plus, HAProxy, Keepalived, cert-manager, Caddy

CI/CD & IaC: GitLab CI, GitHub Actions, Terraform, Ansible, Cronicle, Jenkins

Observability: Prometheus, Grafana, SigNoz, OpenTelemetry, ELK / Elasticsearch, OpenSearch, InfluxDB, Telegraf, Percona PMM, Uptime Kuma, cAdvisor

Databases: Redis / Valkey (Sentinel, HA), PostgreSQL, MySQL, MongoDB, ClickHouse — replication, clustering, backup & restore

AIOps / LLM-Ops: AI-driven observability & incident response — kagent (in-cluster agents), Robusta + HolmesGPT, Robusta KRR (right-sizing), self-hosted LLM agents, log pattern mining (Drain3), z-score anomaly detection & automated root-cause analysis

Messaging & Streaming: Apache Kafka (KRaft), RabbitMQ

Storage: Ceph, MinIO (S3), NFS, LVM / ZFS

Virtualization & Cloud: Proxmox VE, Hyper-V, VMware, Apache CloudStack, EVE-NG, AWS (EC2, S3, CloudFront), DigitalOcean

OS & Networking: Ubuntu, CentOS, Windows Server, Active Directory, DNS/DHCP, pfSense, MikroTik, VLAN, VPN (Tailscale, GRE, WireGuard), firewall & routing

Programming & Scripting: Python, Bash, JavaScript (Node.js, React), SQL, .NET (MVC), Django / Flask

PROFESSIONAL EXPERIENCE

Assistant DevOps Engineer

Onnorokom Projukti Ltd.

Nov 2025 – Present

Dhaka, Bangladesh

Kubernetes & GitOps

- Built and operate multiple high-availability K3s clusters (up to 8 nodes: 3 embedded-etcd control-plane + 5 workers) with MetalLB (L2) load balancing and Envoy Gateway (Gateway API) wildcard-TLS ingress, replacing default Traefik/ServiceLB.
- Implemented GitOps continuous delivery with ArgoCD (auto-sync, self-heal, prune, backup/restore DR), reducing Git-to-cluster reconciliation from 3 minutes to 30 seconds.

- Standardized multi-app / multi-environment deployments using Kustomize base and overlays with ConfigMap-hash auto-rollout; automated wildcard TLS with cert-manager + Let's Encrypt (Cloudflare DNS-01).
- Deployed Istio service mesh (3-replica ingress gateway, Kiali/Prometheus) and secured cluster access with x509 client-certificate authentication and group-based RBAC.
- Hardened cluster reliability with PriorityClass/preemption and PodDisruptionBudget policies, validated on a dedicated test cluster before production rollout.
- Benchmarked ingress with Fortio/iperf3 — sustained 47,000+ QPS at 6 ms p99 and cut TCP retransmits ~100x by re-homing the LoadBalancer VIP onto the client L2 segment.

Databases & High Availability

- Architected 3-node Redis Sentinel + Keepalived/HAProxy HA clusters (quorum 2/3, VIP auto-failover) and reduced master-failover write downtime ~10x (approx. 5s to ~500ms) via HAProxy on-marked-down, with zero application code changes.
- Migrated 10M+ Redis keys cross-cluster in ~5 minutes (~45K ops/sec) using RedisShake, resolving a Redis 8.x stream-encoding incompatibility.
- Operated a 1-shard x 3-replica ClickHouse cluster; implemented native ON CLUSTER backup/restore to MinIO (S3) and cloned/restored ~12M-row databases with verified row-count parity across replicas.
- Deployed a 3-node MongoDB 8.0 replica set (Ansible-provisioned, KeyFile auth) and performed a PostgreSQL 17 physical (byte-level) restore with WAL replay; automated parallel backups and slow-query monitoring.
- Self-hosted Supabase (11-service Docker stack, Kong TLS, Postgres RLS) with automated nightly backups (14-day retention) mirrored offsite to MinIO.

Observability & Logging

- Eliminated an Elasticsearch mapping explosion with Painless ingest pipelines and dynamic:false index templates, cutting field counts from 7,611 to ~25 (and 305K+ to ~30) and removing multi-minute / hanging Kibana Discover loads.
- Performed a zero-downtime migration of a 2.6M-document / 1.1 GB live production index to date-based indexing via an atomic alias swap (no application changes); automated 7–60 day retention with ILM.
- Built full-stack observability with SigNoz + OpenTelemetry on k3s (traces, metrics, logs) and multi-dimensional Prometheus/PromQL alerting across production clusters and 7+ namespaces.
- Authored 3 custom Python Prometheus exporters (Apache mod_status, NGINX access-log RPS, per-container TCP connections); deployed Grafana, Telegraf/InfluxDB, Percona PMM, and Uptime Kuma with Telegram alerting.
- Delivered license-free real-time error alerting to Telegram (ElastAlert2 + Grafana) on a Basic-license cluster, avoiding Platinum-tier licensing cost.

AI-Ops (LLM-Driven Observability)

- Built an in-cluster AI-Ops platform on k3s with kagent — LLM-backed agents (Ollama Cloud) behind Envoy Gateway (TLS + auth) that triage workloads and run live PromQL against Prometheus for real-number answers instead of model guesses.
- Engineered self-hosted "Hermes" AI agents performing statistical (z-score) anomaly detection natively in Prometheus and InfluxDB Flux, auto-correlating each alert with ELK error logs into a root-cause report on Telegram (read-only RBAC); present results in weekly AI-Ops demos.
- Built log pattern-mining (Drain3-derived templates as an ingest-time classification pipeline) that reduced 346,155 raw log documents to 378 distinct patterns at 99.9% coverage — delivering Platinum-tier pattern analytics on a Basic licence at no cost, with a Kibana dashboard and hash-based novelty alerting for new error types.
- Introduced agentless Kubernetes resource right-sizing with Robusta KRR (read-only, Prometheus-driven), surfacing over- and under-provisioned CPU/memory requests across clusters.

Virtualization, Storage & Automation

- Migrated production VMs (Windows Server 2022, Ubuntu) from Hyper-V to Proxmox VE 9 via native VHDX import, engineering a VirtIO/SATA boot fallback that resolved INACCESSIBLE_BOOT_DEVICE BSOD failures.
- Built reusable Proxmox golden templates (Ubuntu cloud-init + Windows Server 2025 sysprep), cutting server provisioning from ~25 minutes to a ~2–3 minute linked clone; automated nightly ZSTD backups with a cross-host DR runbook.
- Deployed 3-node Ceph (Squid) clusters via cephadm serving RBD storage to CloudStack, plus TLS-secured MinIO S3 object storage.
- Ran Apache Kafka 4.2 (KRaft) with end-to-end monitoring (JMX + kafka_exporter, 21-panel Grafana, 6,000+ metrics) and secured RabbitMQ behind an NGINX TLS reverse proxy.

- Operated a self-hosted Cronicle job scheduler processing ~151,000 jobs/day and resolved a production inode-exhaustion outage with retention policies and cleanup automation; managed Harbor private registry (Trivy scanning) and self-hosted GitLab CI/CD.

System Administrator

2022 – 2024

Tickets4Travel

On-site, Dhaka, Bangladesh

- Administered and secured Linux (Ubuntu, CentOS) and Windows Server environments — Active Directory, DNS, DHCP, and group policies.
- Deployed and maintained virtualization platforms (VMware, Hyper-V) for internal infrastructure.
- Managed cPanel/hosting services with automated backup strategies, SSL, and DNS record management.
- Configured and troubleshoot LAN/Wi-Fi networks, routers, NAT, and VPN solutions (PPTP, GRE, Tailscale).
- Monitored system health and authored documentation for IT infrastructure, security policies, and disaster-recovery procedures.
- Supported developers and internal teams by maintaining CI/CD pipelines and server deployments.

PROJECTS

Overengineered To-Do App (DevOps Playground)

- React (Cloudflare Pages + AWS CloudFront/S3), Python backend, PostgreSQL, Redis, ElasticMQ.
- CI/CD with GitHub Actions: lint, test, Docker image scanning, multi-platform builds, automated EC2 deploys, and cache invalidation.
- Repos: github.com/zeeglyismail/todo-frontend · [todo-backend](https://github.com/zeeglyismail/todo-backend) · [ICDevOps](https://github.com/zeeglyismail/todo-frontend) · [ICDevOps](https://github.com/zeeglyismail/todo-backend)

Self-Hosted Open-Source AI (Ollama, Fooocus, Chatterbox)

- Hosted and exposed local LLMs (ChatGPT-style) using Docker, WSL, and Ngrok; demonstrated AI-infrastructure deployment.

CI/CD Deployment with Firebase

- A single GitHub master-branch push triggers automated build and deployment to Firebase Hosting via GitHub Actions.

Full-Stack CRUD Applications

- .NET MVC (4.8 & 7) Result Management System; Django & Flask Visa Management System — RESTful routing, secure authentication, RBAC.
- Repo: github.com/zeeglyismail/sixthSemesterRPS

DineSmart — Dynamic Food-Ordering Web App

- React + Django + PostgreSQL with real-time price calculation on dynamic user input; Dockerized with a CI/CD pipeline.

EDUCATION

B.Sc. in Computer Science & Engineering — Bangladesh Institute of Science & Technology (CGPA 3.17) 2023

HSC (Science) — Nikunja Model College (GPA 3.83) 2019

SSC (Science) — Govt. Science College Attached High School (GPA 4.68) 2017

CERTIFICATIONS

- DevOps — Interactive Cares
- System Administration & IT Infrastructure Services — Google (Coursera)
- Cisco Certified Network Associate (CCNA 101) — CSL
- IT Essentials (Windows, Linux, Virtualization & Cloud) — CSL
- MikroTik & FTTx — CSL
- Cisco Packet Tracer — Udemy
- Cyber Security (Account Cracking) — SystemErrorBD
- WordPress — Udemy

SOFT SKILLS

Adaptability · Communication · Problem-Solving · Quick Learner · Teamwork · Work Ethic · Curiosity · Commitment